

YUANMOU, China

WMO: 567630

Lat: 25.7224N

Lon: 101.8717E

Elev: 3675

StdP: 12.85

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	41.4	43.9	23.7	20.5	75.2	26.5	23.4	74.4	16.9	72.3	15.5	73.2	1.1	90	0.298

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	15.4	96.9	66.9	94.6	67.0	92.6	67.3	74.7	84.3	73.8	83.6	73.2	83.0	8.3	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.6	138.9	77.9	71.7	134.6	77.2	70.9	130.9	76.8	41.3	84.4	40.4	83.7	39.7	83.0	84.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
(4) 15.7	13.7	12.0	DB	36.5	100.3	2.4	2.6	34.8	102.2	33.3	103.7	32.0	105.2	30.2	107.1	
(5)			WB	35.4	77.2	2.2	1.6	33.8	78.3	32.5	79.3	31.3	80.2	29.7	81.4	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	72.3	60.2	65.0	70.8	76.9	80.1	80.8	79.4	79.1	77.1	72.9	65.3	59.7
	DBStd	9.04	4.45	5.00	4.99	5.50	6.75	5.57	4.07	3.66	5.10	4.88	4.16	4.25
	HDD50	5	2	1	0	0	0	0	0	0	0	0	0	2
	HDD65	458	157	55	14	4	3	0	0	0	2	6	46	171
	CDD50	8141	317	421	646	807	934	923	911	902	812	709	458	301
	CDD65	3119	7	55	194	361	471	473	447	437	364	251	53	6
	CDH74	30463	192	736	2164	4024	5460	5102	3926	3631	2853	1709	532	134
	CDH80	12524	11	147	762	1920	2777	2355	1514	1372	1065	518	77	6
(14) Wind	WSAvg	4.5	4.9	5.9	6.0	6.0	5.5	4.9	3.5	3.0	3.3	3.7	3.5	4.0
(15) Precipitation	PrecAvg	24.4	0.2	0.1	0.2	0.5	1.6	4.5	5.6	4.9	3.6	2.2	0.8	0.3
	PrecMax	35.7	0.7	0.5	1.1	2.8	4.8	9.0	11.9	12.3	7.4	9.1	3.7	1.7
	PrecMin	11.3	0.0	0.0	0.0	0.0	0.0	0.6	1.8	2.3	1.2	0.1	0.0	0.0
	PrecStd	4.9	0.2	0.2	0.3	0.7	1.3	2.1	2.3	2.1	1.7	1.8	0.9	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.4	87.4	91.7	97.2	100.8	100.1	95.7	94.7	93.9	89.8	85.3	80.7
		MCWB	54.8	57.1	59.0	62.6	67.0	70.0	71.4	71.7	70.8	67.7	61.4	55.7
	2%	DB	78.5	84.2	89.1	94.4	97.4	96.1	92.5	91.9	91.2	87.5	82.2	77.5
		MCWB	53.5	55.9	58.3	61.6	65.9	70.2	72.1	71.8	70.5	66.6	60.7	55.4
	5%	DB	76.1	81.5	86.9	92.3	94.8	93.1	90.2	89.6	88.9	85.5	79.8	75.0
		MCWB	53.2	55.0	57.6	61.1	65.8	70.6	72.2	71.6	70.0	66.2	60.0	54.7
	10%	DB	73.4	78.9	84.6	89.9	92.2	90.6	87.8	87.3	86.4	83.1	77.2	72.1
		MCWB	52.4	54.1	57.0	60.7	65.6	70.7	72.1	71.5	69.6	65.7	59.2	54.5
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	57.5	59.7	62.0	65.9	72.4	75.5	76.2	75.6	73.8	70.8	66.6	59.8
		MCDB	75.0	82.1	83.3	86.0	85.6	85.9	85.5	85.3	83.3	81.0	75.5	73.3
	2%	WB	55.7	57.5	60.3	64.2	70.8	74.1	74.8	74.3	72.7	69.6	64.4	57.6
		MCDB	73.3	79.6	82.0	86.5	84.3	84.9	84.1	83.8	83.0	80.1	74.9	71.9
	5%	WB	54.5	56.1	59.3	63.2	69.7	73.2	74.0	73.5	71.9	68.8	62.8	56.3
		MCDB	71.6	77.7	81.0	85.3	83.3	84.3	83.3	83.0	82.2	79.3	73.7	70.6
	10%	WB	53.4	54.9	58.2	62.2	68.7	72.5	73.3	72.8	71.1	67.9	61.1	55.1
		MCDB	69.9	75.9	79.9	84.0	83.3	84.0	82.5	82.0	81.3	78.3	72.6	69.1
(35) Mean Daily Temperature Range	5% DB	MDBR	27.2	27.3	26.1	23.1	18.7	15.4	14.1	15.5	15.4	17.6	23.2	25.4
		MCDBR	31.1	30.9	29.5	26.3	22.7	19.0	18.4	19.6	19.1	21.5	27.4	29.5
		MCWBR	11.2	10.1	8.6	6.7	5.4	5.0	4.8	4.7	4.3	5.4	8.7	10.7
	5% WB	MCDBR	28.2	28.5	25.2	22.7	17.4	15.4	14.9	16.0	15.9	16.2	21.7	25.7
		MCWBR	11.1	10.0	8.0	6.4	5.7	5.8	5.2	4.9	4.6	4.9	7.9	10.3
(40) Clear-Sky Solar Irradiance	taub		0.339	0.357	0.499	0.501	0.457	0.435	0.455	0.496	0.469	0.430	0.377	0.360
	taud		2.398	2.355	1.922	1.994	2.222	2.337	2.266	2.124	2.185	2.265	2.340	2.347
	Ebn at Noon		290	292	254	257	268	272	266	255	260	264	273	276
	Edh at Noon		33	37	60	58	46	41	43	50	46	40	35	34
(44) All-Sky Solar Radiation	RadAvg		1318	1604	1737	1812	1708	1510	1397	1403	1244	1183	1276	1165
	RadStd		108	152	161	165	205	157	99	121	108	146	147	104

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.83	N/A	N/A	+1.78	-0.34	-0.58	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		+0.70	N/A	N/A	+1.48	N/A	-0.40	N/A	N/A	N/A	N/A

Nomenclature: See separate page